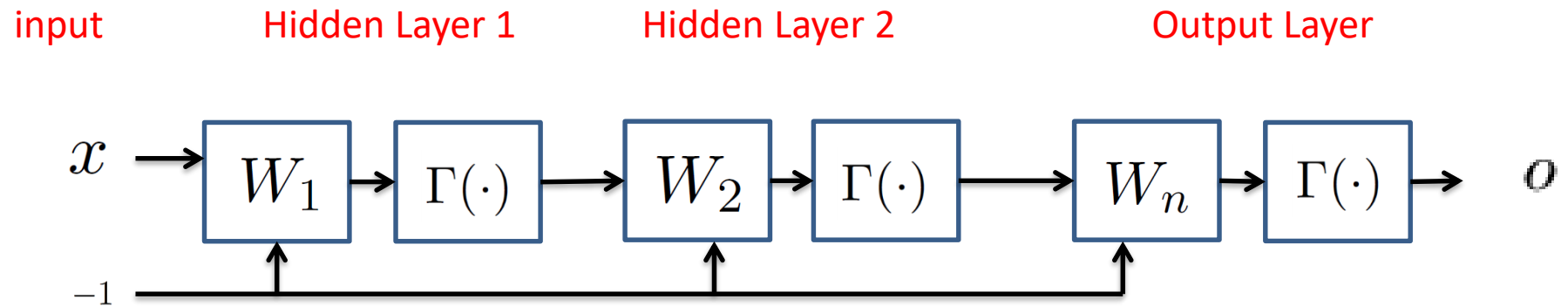


Gradient-Based Optimization

1. D. Bertsekas, *Convex Analysis and Optimization*, Athena Scientific (2003).
2. S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge Univ. Press (2004).
3. Yu. Nesterov, *Introductory Lectures on Convex Optimization*, Springer (2004).
4. J. Nocedal and S. Wright, *Numerical Optimization*, Springer (1999).
5. G. Strang, *Linear Algebra and Learning from Data*, Cambridge, 2019.



$$o = F(W, x)$$

$$\mathcal{C}_1 \rightarrow d_1 \quad , \quad \mathcal{C}_2 \rightarrow d_2$$

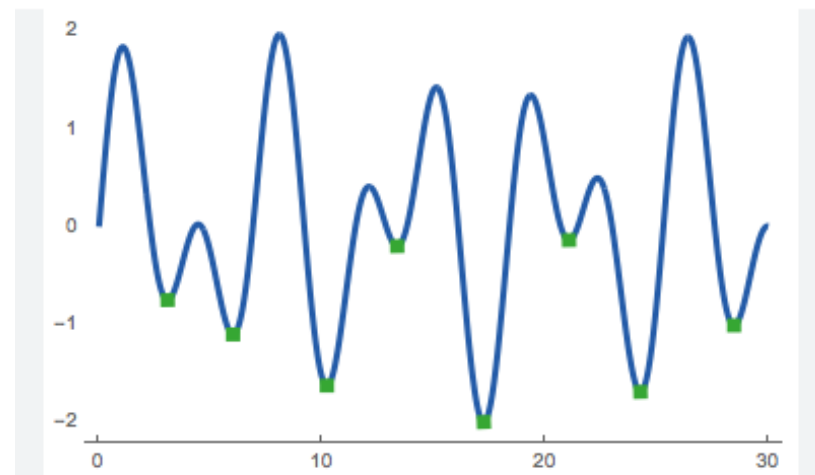
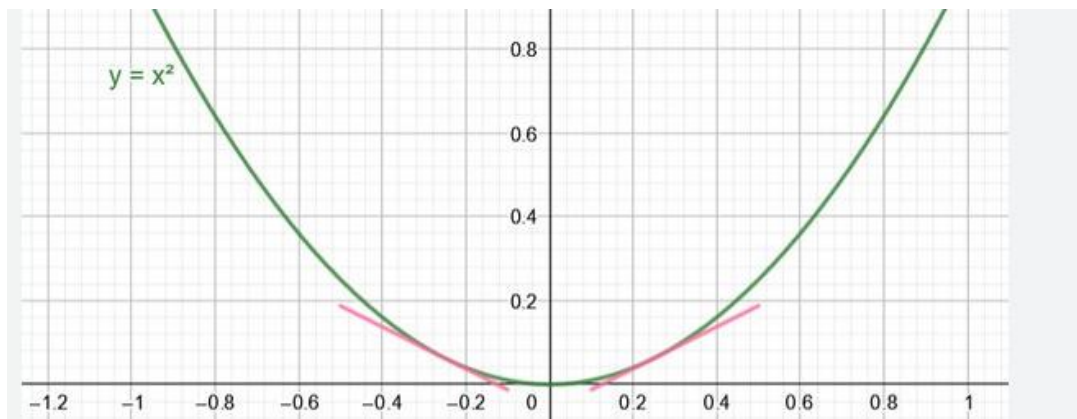
$$x \in \mathcal{C}_1 \rightarrow error = d_1 - F(W, x)$$

$$x \in \mathcal{C}_2 \rightarrow error = d_2 - F(W, x)$$

Total Error : $E(W) = \sum_{x \in \mathcal{C}_1} \|d_1 - F(W, x)\| + \sum_{x \in \mathcal{C}_2} \|d_2 - F(W, x)\|$

Generic problem : $\min_x f(x)$

$$\boldsymbol{x}^* = \arg \min f(\boldsymbol{x})$$



• How to find a minimum ?

• **How to find a minimum ?**

$$f'(x) = 0$$

• How to find a minimum ?

$$f'(x) = 0$$

not feasible in our case

$$f : \mathbf{R}^n \rightarrow \mathbf{R}$$

$$\nabla f(x_1, \dots, x_n) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

gradient



Iterative Algorithm

Start with x_0

Find x_1 such that $f(x_1) \leq f(x_0)$

$\vdots$ $\vdots$

Find x_{k+1} such that $f(x_{k+1}) \leq f(x_k)$

Iterative Algorithm

Start with x_0

Find x_1 such that $f(x_1) \leq f(x_0)$

$\vdots$ $\vdots$

Find x_{k+1} such that $f(x_{k+1}) \leq f(x_k)$

$$x_{k+1} = x_k + \Delta x_k$$

$$\Delta x_k = ?$$

How to choose Δx_k ?

Gradient Descent (First Order Method)

$$f(x) = f(x_0) + (x - x_0)^T \nabla f(x_0) + h.o.t.$$

$$x - x_0 = -\mu \nabla f(x_0) \quad \Rightarrow \quad x = x_1 = x_0 - \mu \nabla f(x_0)$$

$$f(x_1) = f(x_0) - \mu \|\nabla f(x_0)\|^2 + h.o.t.$$

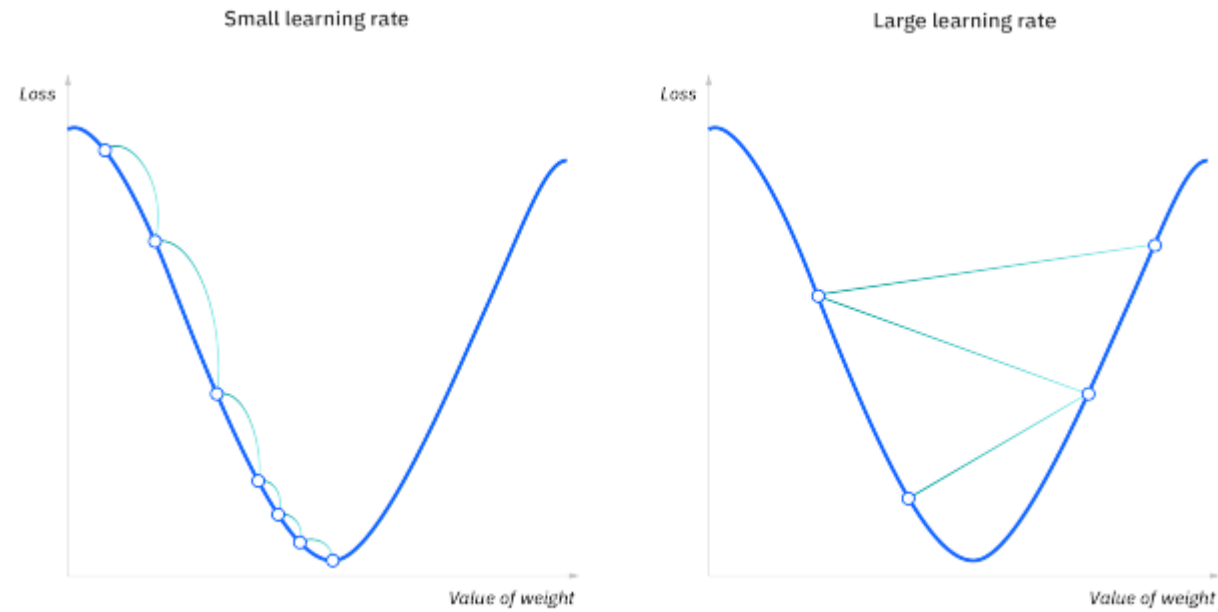
If h.o.t. is negligible

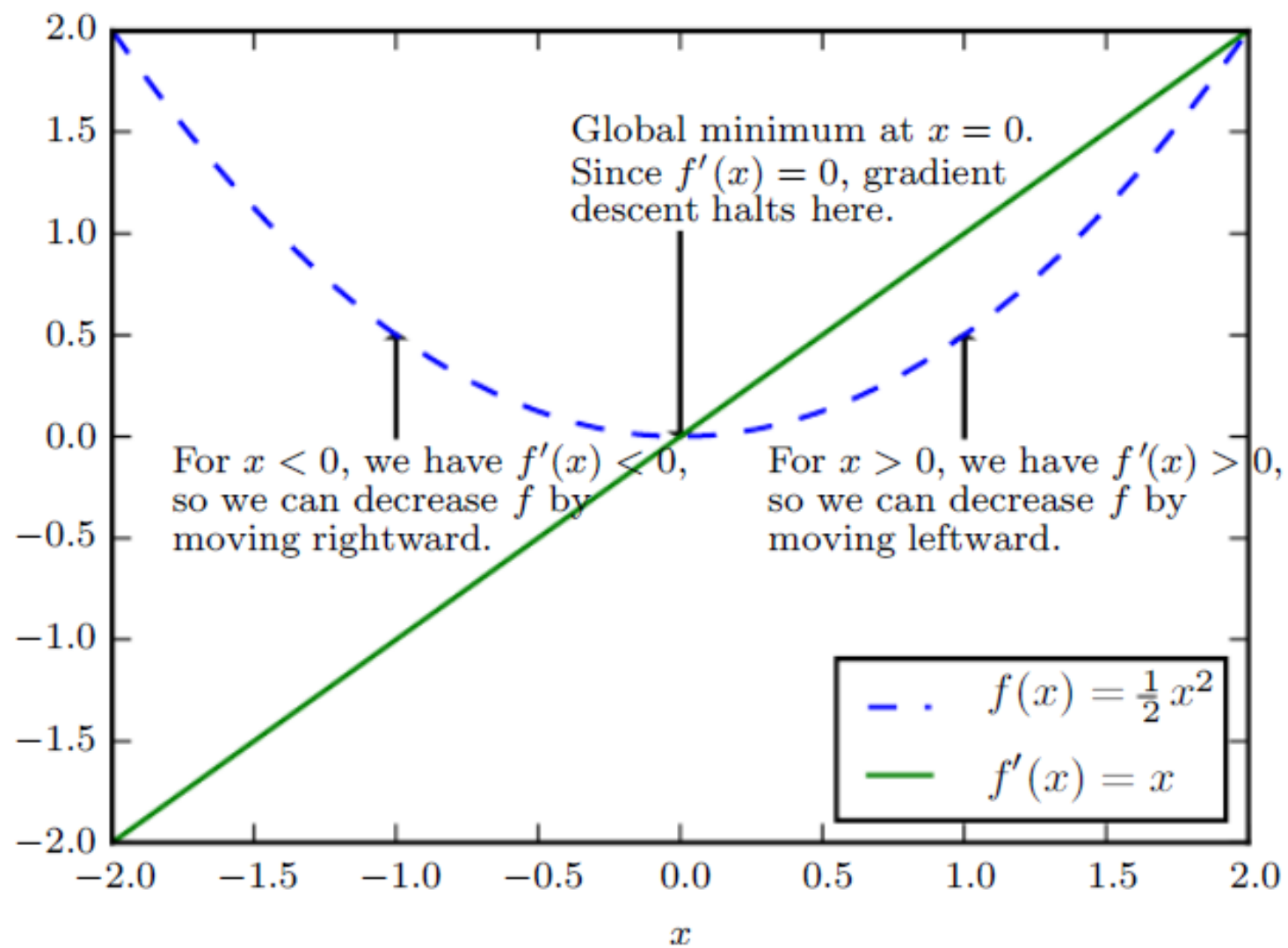
$$f(x_1) < f(x_0) \quad \text{when } \nabla f(x_0) \neq 0$$

Gradient Descent Algorithm :

$$x_{k+1} = x_k - \mu \nabla f(x_k)$$

If converges $\Rightarrow x_{k+1} = x_k \Leftrightarrow \nabla f(x_k) = 0 \Rightarrow$ minimum.





What if we choose other search directions ?

$$x_{k+1} = x_k + \mu v \quad v \neq -\nabla f ?$$

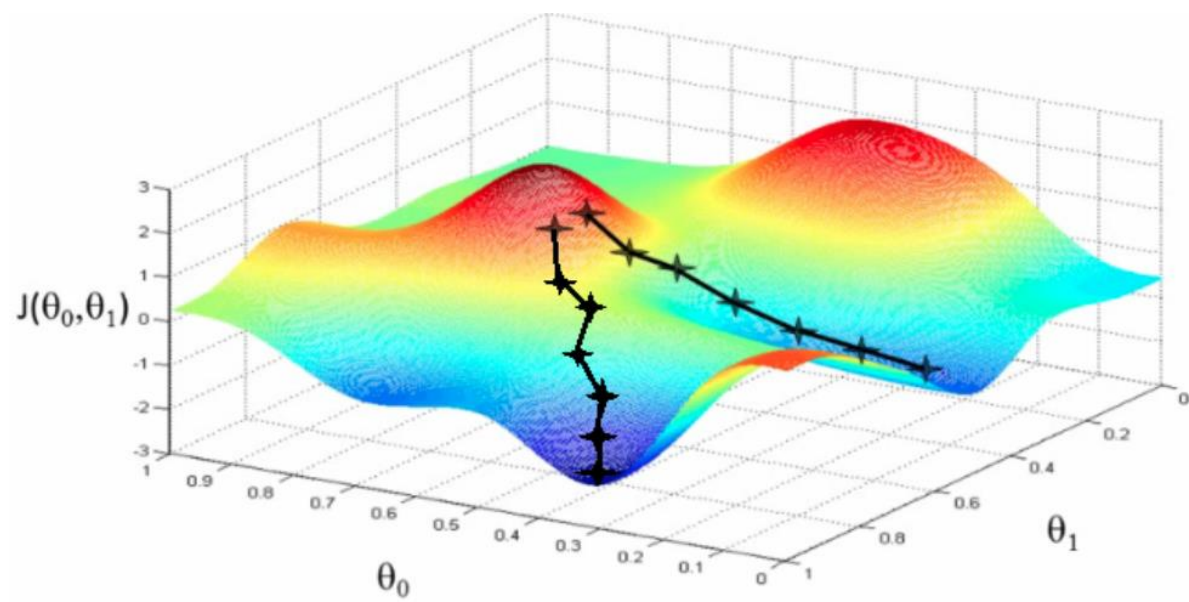
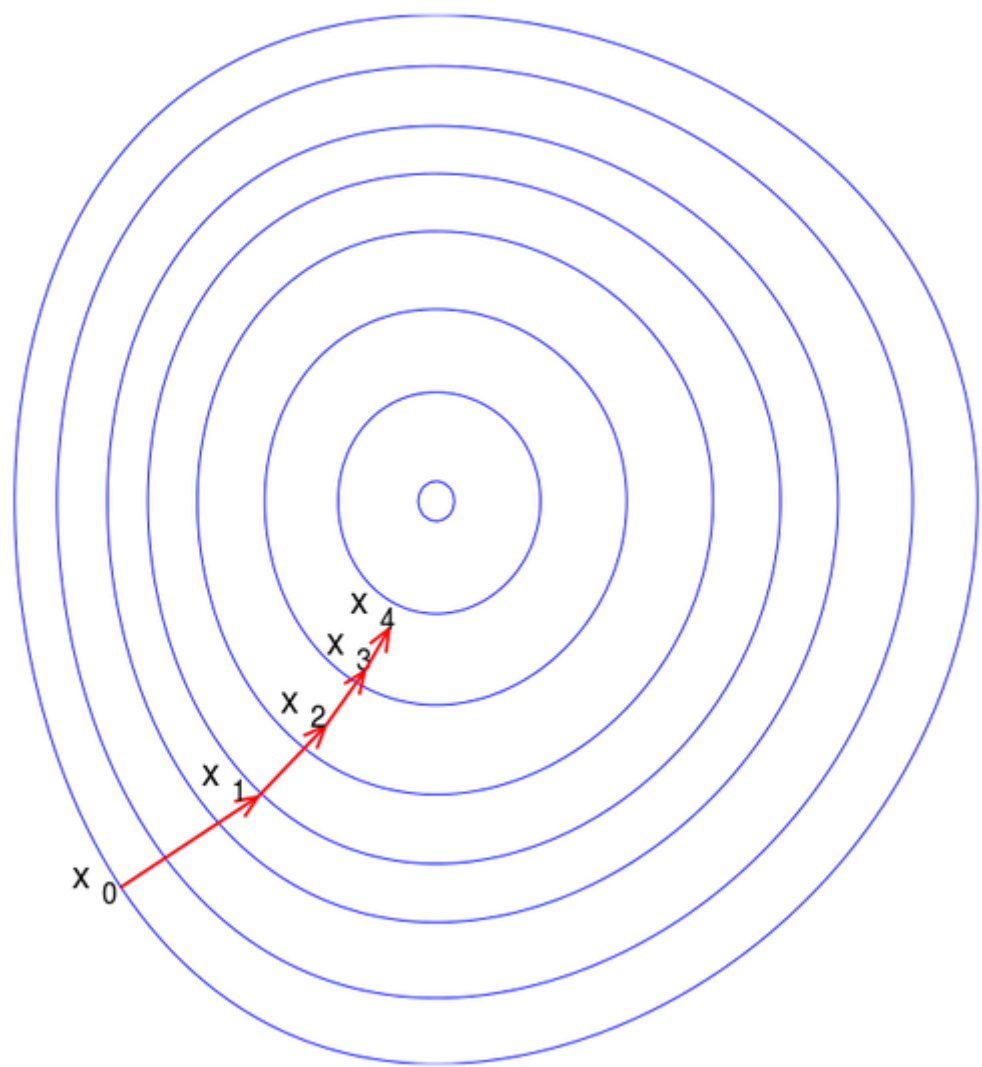
$$\nabla_{\mathbf{v}} f(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{v}) - f(\mathbf{x})}{h}.$$

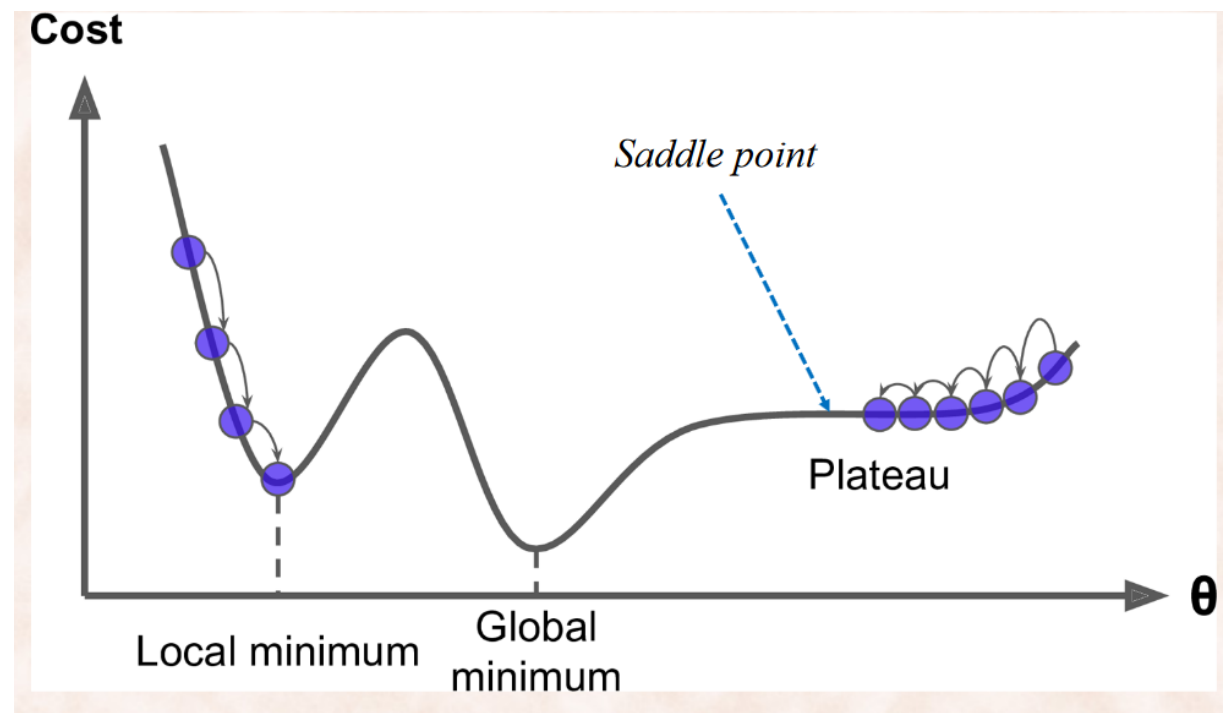
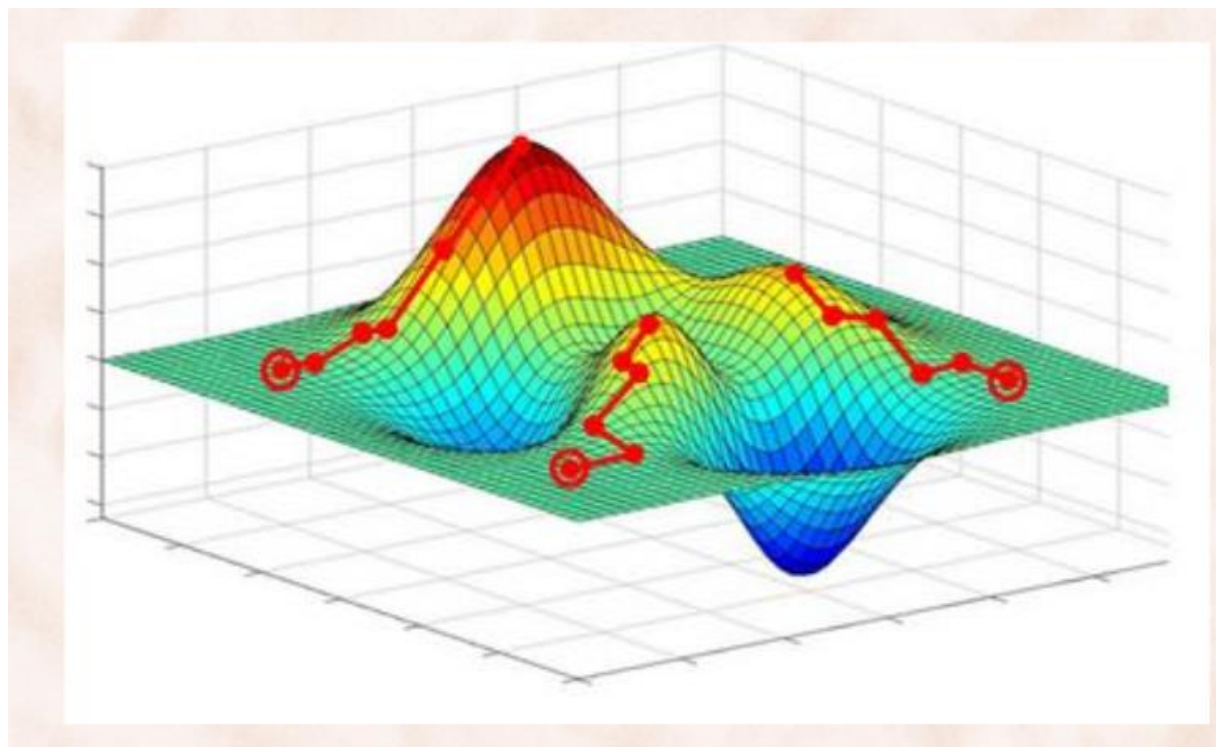
https://en.wikipedia.org/wiki/Directional_derivative

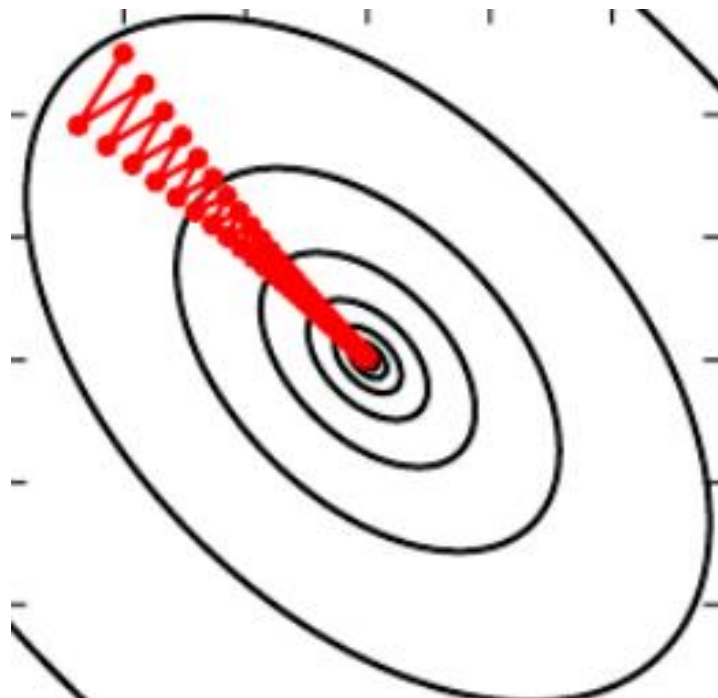
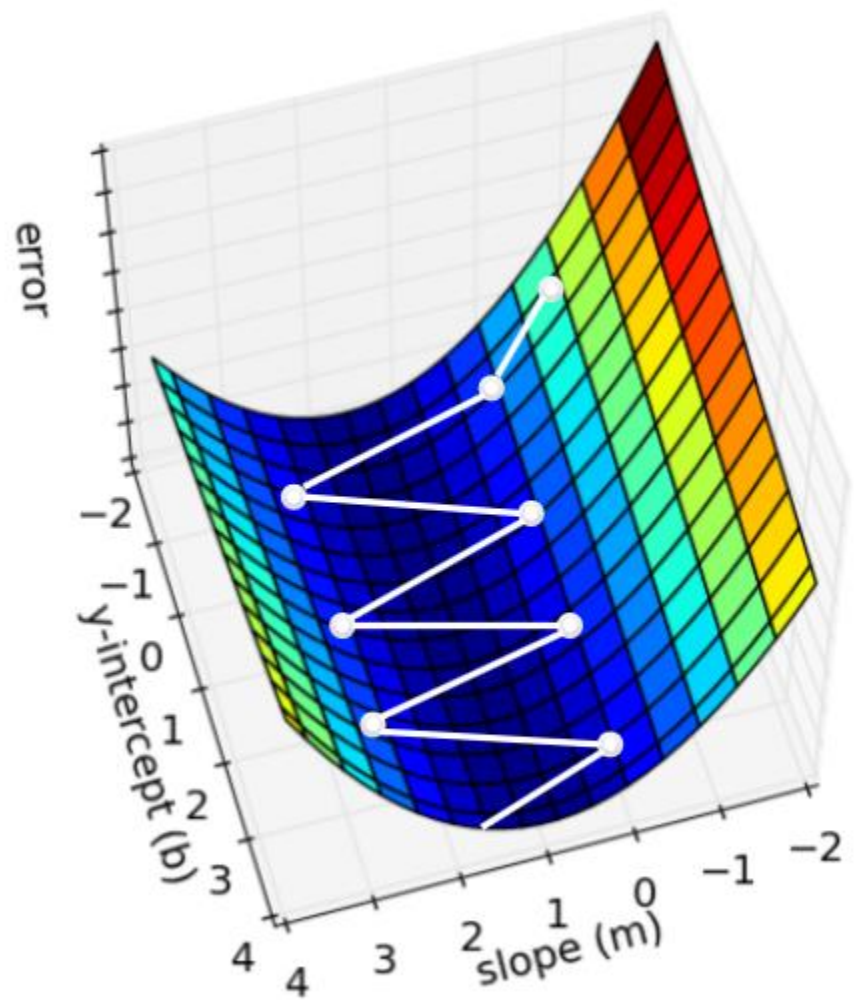
$$\nabla_{\mathbf{v}} f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{v}$$

$$f(x + hv) \approx f(x) + h \nabla f \cdot v \quad \|\nabla f\| \cdot \|v\| \cos(\theta)$$

Decrease is most negative when $\theta = \pi \Rightarrow v = -\mu \nabla f$







Hessian matrix. The Hessian matrix $\mathbf{H}(f)(\mathbf{x})$ is defined such that

$$\mathbf{H}(f)(\mathbf{x})_{i,j} = \frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{x}).$$

$$\frac{\partial^2}{\partial x_i \partial x_j} f(\mathbf{x}) = \frac{\partial^2}{\partial x_j \partial x_i} f(\mathbf{x}).$$

This implies that $H_{i,j} = H_{j,i}$, so the Hessian matrix is symmetric

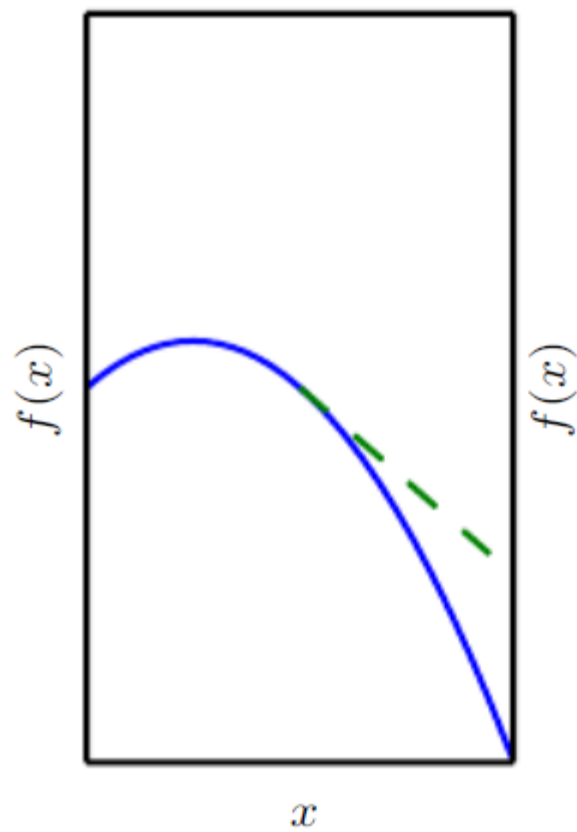
$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^{(0)}) + (\boldsymbol{x} - \boldsymbol{x}^{(0)})^\top \boldsymbol{g} + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^{(0)})^\top \boldsymbol{H}(\boldsymbol{x} - \boldsymbol{x}^{(0)}).$$

for gradient descent $\boldsymbol{x} = \boldsymbol{x}(0) - \epsilon \boldsymbol{g}$

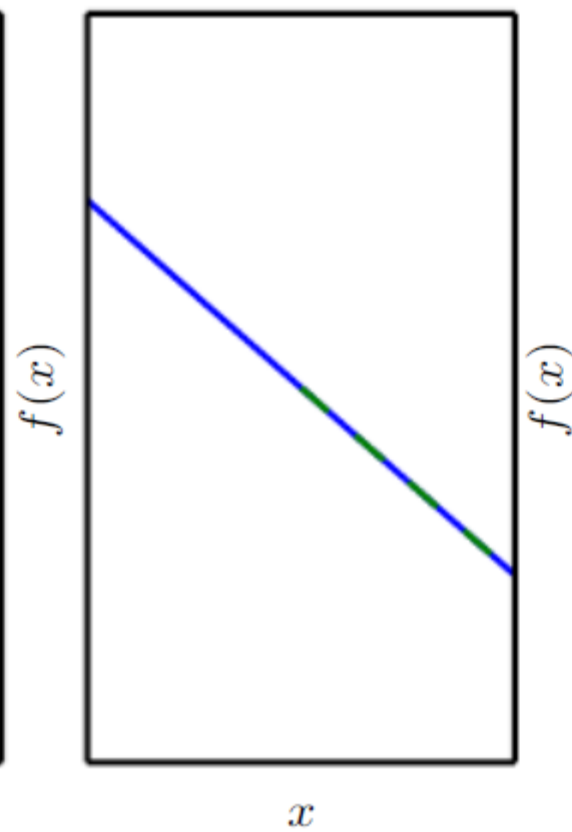
$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^{(0)}) - \epsilon \boldsymbol{g}^\top \boldsymbol{g} + \frac{1}{2} \epsilon^2 \boldsymbol{g}^\top \boldsymbol{H} \boldsymbol{g}.$$

$$\epsilon^* = \frac{\boldsymbol{g}^\top \boldsymbol{g}}{\boldsymbol{g}^\top \boldsymbol{H} \boldsymbol{g}}.$$

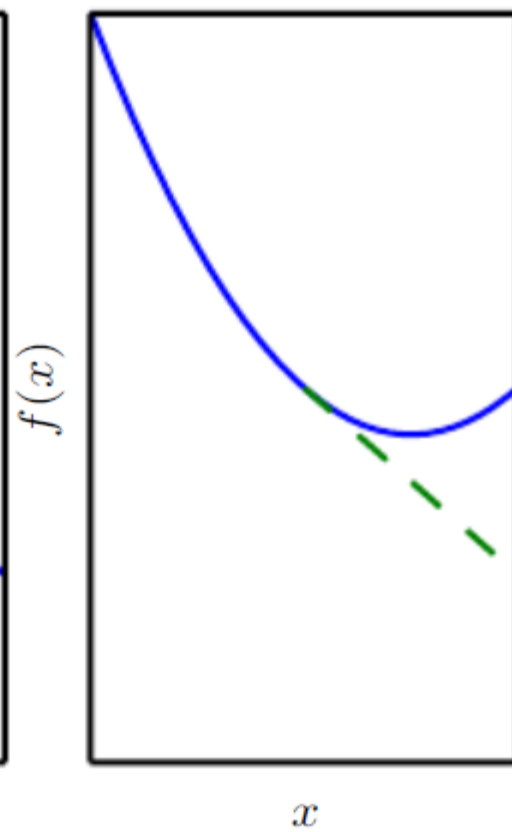
Negative curvature



No curvature



Positive curvature



Newton's method.

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^{(0)}) + (\boldsymbol{x} - \boldsymbol{x}^{(0)})^\top \nabla_{\boldsymbol{x}} f(\boldsymbol{x}^{(0)}) + \frac{1}{2} (\boldsymbol{x} - \boldsymbol{x}^{(0)})^\top \boldsymbol{H}(f)(\boldsymbol{x}^{(0)}) (\boldsymbol{x} - \boldsymbol{x}^{(0)}).$$

$$\text{Generic problem} \quad : \quad \min_x f(x)$$

$$\nabla f(x) = \nabla f(x_0) + H(f)(x - x_0) = 0$$

$$\boldsymbol{x}^* = \boldsymbol{x}^{(0)} - \boldsymbol{H}(f)(\boldsymbol{x}^{(0)})^{-1} \nabla_{\boldsymbol{x}} f(\boldsymbol{x}^{(0)}).$$

Newton's method.

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^{(0)}) + (\boldsymbol{x} - \boldsymbol{x}^{(0)})^\top \nabla_{\boldsymbol{x}} f(\boldsymbol{x}^{(0)}) + \frac{1}{2} (\boldsymbol{x} - \boldsymbol{x}^{(0)})^\top \boldsymbol{H}(f)(\boldsymbol{x}^{(0)}) (\boldsymbol{x} - \boldsymbol{x}^{(0)}).$$

Generic problem : $\min_{\boldsymbol{x}} f(\boldsymbol{x})$

$$\nabla f(\boldsymbol{x}) = \nabla f(\boldsymbol{x}_0) + \boldsymbol{H}(f)(\boldsymbol{x} - \boldsymbol{x}_0) = 0$$

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$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \boldsymbol{H}(\boldsymbol{x}_k)^{-1} \nabla f(\boldsymbol{x}_k)$$

Example: Linear Least Squares

Example Problem : $Ax = b$ when A^{-1} does not exist

Least Squares Solution $\min_x \|Ax - b\|_2$

(Moore-Penrose) Pseudo-inverse $x^* = A^+b$

$$f(\mathbf{x}) = \frac{1}{2} \|\mathbf{Ax} - \mathbf{b}\|_2^2$$

$$\mathbf{x}^* = \arg \min f(\mathbf{x}) \quad =?$$

$$\text{If } A^{-1} \text{ exists} \quad \Rightarrow \quad x^* = A^{-1}b$$

$$\|y\|^2 = y^T y \quad \Rightarrow \quad f(x) = \frac{1}{2} \|Ax - b\|^2 = \frac{1}{2} (Ax - b)^T (Ax - b)$$

$$\nabla_x f(x) = A^T (Ax - b) = A^T Ax - A^T b$$

$$\text{If } (A^T A)^{-1} \text{ exists} \quad \Rightarrow \quad x^* = (A^T A)^{-1} A^T b$$

Algorithm 4.1 An algorithm to minimize $f(x) = \frac{1}{2} \|Ax - b\|_2^2$ with respect to x using gradient descent, starting from an arbitrary value of x .

Set the step size (ϵ) and tolerance (δ) to small, positive numbers.

while $\|A^T Ax - A^T b\|_2 > \delta$ **do**

$x \leftarrow x - \epsilon (A^T Ax - A^T b)$

end while

What if $A^T A$ is not invertible ??

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$$x^* = A^\dagger b$$

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$$\hat{f}(x) = \frac{1}{2} \|Ax - b\|^2 + \frac{\lambda}{2} x^T x$$

$$\hat{x}^* = \arg \min_x \hat{f}(x)$$

What if $A^T A$ is not invertible ??

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$$\hat{x}^* = (A^T A + \lambda I)^{-1} A^T b$$

Newton Method and Hessian

$$f(x) = \frac{1}{2} \|Ax - b\|^2 = \frac{1}{2} (Ax - b)^T (Ax - b)$$

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Newton Method and Hessian

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$$H = A^T A$$

Newton Method and Hessian

$$f(x) = \frac{1}{2} \|Ax - b\|^2 = \frac{1}{2} (Ax - b)^T (Ax - b)$$

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$$x_{k+1} = x_k - H(x_k)^{-1} \nabla f(x_k)$$

Newton Method and Hessian

$$f(x) = \frac{1}{2} \|Ax - b\|^2 = \frac{1}{2} (Ax - b)^T (Ax - b)$$

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$$x_{k+1} = x_k - H(x_k)^{-1} \nabla f(x_k)$$

$$x_{k+1} = (A^T A)^{-1} A b$$